Unit 4 Mini-Cases

Principles of Capital Markets

Winter Term 2025/26

Please upload your solutions and indicate which Mini-Cases (MCs) you have solved before **November 12, 2025 at 6am**. To do so, go to the assignment section on Canvas and choose "Unit 4 Indication and Submission of MCs". Then answer the true/false questions (Questions 1 to 6) and upload your solutions file (Question 7). **There will be no deadline extensions! Only complete submissions containing answers to the T/F questions and a solutions upload will be considered, your last complete attempt counts!** You can either write your solutions using software or scan your handwritten solutions. Please note that you may only submit one solutions file. If your solutions consist of more than one file (e.g. 1 pdf and 1 Excel file), please upload a .zip file. Please make sure to hand in only one file of each file type (you might have to merge your scans to one pdf file). Uploaded files will be checked for plagiarism. The solutions in your file should be self-explanatory - not only for the instructor, but for other students as well. Indicate only Mini-Cases you feel confident to present.

<u>Important remark for the presentations:</u> Make sure to provide a concise explanation of your solutions. In particular, when applying any formula from the lecture materials or other sources, you may be asked to explain the structure of your solutions and provide the intuition and underlying concepts behind the formula used.

Mini-Case 4.1 (Excel or R)

Use the following information on the annualized mean returns and covariances for the asset classes GOLD, STOCKS, REIT and BONDS which you have already encountered in Unit 3. Assume that historical data provides unbiased estimates of future performance.

	Correlation with					
Asset	GOLD	STOCKS	REIT	BONDS	Mean return (%)	Standard deviation (%)
GOLD	1	0.10	0.20	0.45	13.14	13.99
STOCKS	0.10	1	0.78	0.12	13.63	15.28
REIT	0.20	0.78	1	0.34	4.26	17.39
BONDS	0.45	0.12	0.34	1	1.24	5.38

You are now considering all four asset classes for your portfolio.

(a) What are the expected return and variance of an equally weighted portfolio consisting of GOLD, STOCKS, REIT and BONDS?

(b) Show that you can improve on the portfolio in part *(a)* by reducing its risk. Calculate the weights of the minimum variance portfolio numerically **by directly minimizing the portfolio variance** using a tool such as Excel Solver.

Mini-Case 4.2 (Excel or R)

Use the information provided in *Mini-Case 4.1* for this exercise. Again, you are considering all four asset classes for your portfolio.

(a) Calculate the weights of the minimum variance portfolio numerically **by applying the marginal variance approach** using a tool such as Excel Solver.

(b) What are the expected return and variance of the minimum variance portfolio?

Mini-Case 4.3 (pdf)

This case contains general questions designed to test your understanding of the mean-standard deviation diagram. Complete the following tasks and explain each step in detail:

(a) Draw a mean-standard deviation diagram that illustrates combinations of a single risky asset and a risk-free asset.

(b) Extend the diagram to show the set of all possible combinations between the risk-free asset and all available risky portfolios.

(c) Clearly label the Capital Market Line (CML), the Minimum Variance Portfolio (V), and the Tangency Portfolio (T) on your diagram.

(d) Explain why the Capital Market Line dominates all other possible portfolio combinations.

(e) What condition must hold at the tangency portfolio, and why?

Hint: Refer to the lecture slide titled "Identification of the Tangency Portfolio."

Mini-Case 4.4 (Excel or R)

Use the data provided in *Mini-Case 4.1* for this exercise.

(a) Compute the weights of the tangency portfolio consisting of all four asset classes. Assume that the risk-free rate is 0.5%.

Hint: Follow the procedure outlined on page 68 of the lecture slides.

(b) Calculate the expected annual return and variance of the tangency portfolio.

Mini-Case 4.5 (Excel or R)

Use the data provided in *Mini-Case 4.1* for this exercise.

(a) Create an efficient frontier diagram for GOLD, STOCKS, REIT and BONDS. Plot both the tangency portfolio and the Capital Market Line (CML) on the same graph. *Hint: You can construct the efficient frontier using combinations of two assets that lie on the frontier, such as the minimum variance portfolio and the tangency portfolio. See page 69 of the lecture slides for guidance.*

(b) Calculate the beta of each asset class (GOLD, STOCKS, REIT and BONDS) with respect to the tangency portfolio obtained in *MC 4.4*.

(c) Calculate the beta of an equally weighted portfolio of GOLD, STOCKS, REIT and BONDS with respect to the tangency portfolio.

Mini-Case 4.6 (Excel or R)

Use the stock data provided in Unit 2.

(a) Construct an equally weighted portfolio of all four stocks. What are the average return and standard deviation of this portfolio?

(b) Assume that the portfolio in *a)* is your market portfolio. Select one stock, i.e. Oracle, and compute the beta of this stock with respect to your market portfolio.

(c) Use the Capital Asset Pricing Model (CAPM) to estimate the expected return of your selected stock, i.e. Oracle. Assume a risk-free rate of 2%.